Research on Driving Factors of COVID-19 Protective Behaviours in Australia Based on Deep Learning

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1 Declaration

In submitting this work I am indicating that I have read the University's Academic Integrity Policy. I declare that all material in this assessment is my own work except where there is clear acknowledgement and reference to the work of others. I give permission for this work to be reproduced and submitted to other academic staff for educational purposes.

2 Abstract

During the COVID-19 pandemic, personal protective behaviour, especially wearing masks, played a crucial role in controlling of the infection. Although public health policies such as mask mandates significantly changed people's compliance, understanding the behavioural driving factors behind protective behaviours are still important. These driving factors are complex and dynamically influenced by macro pandemic factors such as vaccination rates and confirmed cases. This study investigates the determinants of COVID-19 protective behaviours in Australia through a two-phase machine learning framework. In the first phase, this study reproduced Ryan et al.'s (Ryan et al. 2025, 8–15) research, using random forest and XGBoost to classify and predict the protective behaviours before and after the mask mandates. The evaluation results showed that prior to the implementation of the mandates, people's protective behaviour was mainly driven by existing protective habits and frequency of non-household contact. After the mandates, the passage of time (week number) and an individual's willingness to isolate became the main predictive factors. To address the structural limitations of traditional tree models in capturing high-order and nonlinear variable interactions, the upcoming second phase proposes the construction of a Multi-layer Perceptron (MLP) architecture and the integration of state-level weekly vaccination rates and confirmed cases. In addition, to overcome the 'black box' characteristics of neural networks, this study introduced SHAP (SHapley Additive exPlanations) values to quantify feature contributions.

3 Introduction

During the COVID-19 pandemic, individual protective behaviours played a crucial role in controlling the spread of the virus (Guo, Xiang, and Wang 2023, 1–2). Among all protective behaviours, mask wearing was initially recognized as one of the most effective measures [Eikenberry et al. (2020), p.11-13). To cope with future pandemics, understanding the drivers of protective behaviours, particularly mask wearing, is essential for the design of public health policies.

There were many studies investigating why people wear masks from various perspectives (Ryan et al. 2025, 2). Among these, mandated policy is one of the most important factors, which may lead to a sharp increase in the use of masks [Haischer et al. (2020), p.4-9). Recently, Ryan et al. (Ryan et al. 2025, 8–13) investigated the role of face mask mandates in influencing the drivers of protective health behaviours in Australia. However, some key factors have not been considered. For example, the introduction of vaccination may alter individuals' risk perception and behavioural responses (De Nicolò, Villari, and De Vito 2022, 2). There is a need to understand how different external factors drive behaviour change to inform future public health policy-making.

In order to identify these complex relationships, appropriate modelling approaches are needed. Most existing research relies on traditional statistical or tree-based models. Although tree-based models often perform strongly on tabular data, they may be limited in fully capturing complex interactions (Rana et al. 2023, 3). However, research also indicates that for most tabular data, GBDTs(gradient-boosted decision trees) outperform neural networks (NNs).In fact, the comparative performance of these two types of models on tabular data remains a subject of significant academic debate, as results often vary significantly across different datasets. Upon reviewing the literature, there is no existing research applying deep learning to similar datasets within this field.

To address these research gaps, this study integrated vaccination data and COVID-19 case numbers, and will apply deep learning method to find out whether it can capture complex relationships and identify the key drivers of protective behaviours, especially mask wearing.

For the whole study, the primary research questions are:

- How do vaccination rates and COVID-19 case numbers influence the uptake of protective behaviours in Australia?
- How does the performance of deep learning models compare to traditional statistical or tree-based models in predicting protective behaviours?

4 Background

5 Methods (or possibly a more informative heading)

This section details the experimental design and implementation methods of the study.

5.1 Data Sources

This section will first provide an overview of the background and cleaning methods of all four datasets, and then detail the modelling strategies corresponding to different stages in sections.

"Does the reader have all the information they need (from the Background and Methods sections) so that they can reproduce my results exactly?"

If you can answer this in the affirmative, then you are probably on the right track.

We can type equations in our new environment:

$$E[\frac{1}{X}] \neq \frac{1}{E[X]} \tag{1}$$

You can see that the text is indented and so it is clearly separated from the normal text because the next line is just normal text.

Here is the normal text outside our Definition environment. Now, how about a theorem?

Theorem (KnuthTextbook?)

This is a theorem that we need. It comes from the earlier work cited in the title of this environment, and we include it here so that the reader has all the information that they need to see what we are doing. As we did above, you can have dot points in your new environment.

1. first result in the theorem
2. second result in the theorem

Now an equation in the theorem

$$b = a + 2 \tag{2}$$

This text is after the theorem.

5.2 Algorithms

You will very likely want to describe some algorithms or processes that you followed in your project. For that, you could use the same structure as above.

Model algorithm

- Step 1: Define the initial values of the parameters n, a, c_1, its
- Step 2: Generate all the random points that we will sample from.
- Step 3: Sample n points from the population.
  - Repeat this its times.

5.3 Something fancy

If you want to really separate something from the main text, you could try putting a box around it with the following code.

Theorem (KnuthTextbook?)

This is the same theorem as above, but now in a facny box!

5.4 Maths equations

You can see above that we've got a couple of equations, which were defined using `\begin{align}<maths>\end{align}`. Note that we also included a label with `\label{<name>}`. This is handy, since it allows us to refer to the equation in our text, eg. from Equation (2) we can see that b is two more than a.

6 Results

Remember the advice about the previous sections. You need your results to be concise, but once again be concrete, quantitative, and provide enough information that the results could be reproduced and verified. Figures and tables can be very useful. However, while a picture is worth a thousand words, this is not true by itself. Any graph or figure included in a report MUST have:

1. a detailed caption describing exactly what the figure shows (it should almost stand alone);
2. appropriate axes with labels including units; and
3. discussion in the text of the document, not just the caption (make sure you refer to exact figure numbers in the text).

Moreover, they should be easy to read with large enough text, and clearly marked data points. Tables are also good and should follow the same rules. To produce a table, you can perhaps print out a tibble using an R code chunk, or you can type the table using Latex commands like the one in Table 1.

cell1	cell2	cell3
cell4	cell5	cell6
cell7	cell8	cell9

Table 1: Table to test captions and labels.

A few figures and tables in a document are very useful, but be aware that deluging a reader with figures and tables can be counter-productive. Part of the art of technical writing is choosing good ways of informing the reader of the critical information without diluting it with volumes of irrelevancies.

IMPORTANT:

- **All of your results must be backed up by some code output.**
  - **Any claims that have no supporting evidence will be ignored!**
- So that means that any claims you make should reference the place in the appendix or some code where you produce the calculations or graphs that you use. You can use a footnote for this or just put a comment in brackets.
  - Here's how you can insert a footnote[1].
- Eg. Likewise, the sons of Feanor, Celegorm and Curufin, dwelt in Nargothrond, and their oath to regain the Silmarils would likewise call them. According to our simulation, on average

[1] Including a footnote

there was space for all these human and elven heroes 97% of the time.[2]

- Eg. He pursued the Orcs and came in secret to their fire, witnessing their cptain holding aloft his father's hand as a trophy, bearing the ring of Felagund. Beren slew the captain and took the hand with the ring, managing to escape before the Orcs could kill him. We also found that the average waiting time experienced by Beren was 5.78 seconds. (See "average waiting time" in Appendix E.2. This calculation was generated by the function `wait_beren()` on lines 57 – 73 of the file `silmarils_fns.R`.)

7 Discussion

Here you discuss the implications of your results in the context of your research project. In other words, tell the reader what you think the results mean. This section will probably involve a bit of a short summary, as you need to remind the reader of what you are doing so that the results are meaningful.

Since research is never finished, you should discuss what the next steps in the project should be.

8 Acknowledgements

This is where should acknowledge any help you got, other than those things listed in the references. So things like:

- Research advice given to you by someone else
- Help provided by an AI agent
  - Some AI help is ok, as long as it doesn't do your work for you and you acknowledge and reference it
  - Ask your supervisor for advice on what AI help is appropriate
- Help given by the Student Advisors

For example, I acknowledge that some of this template was adapted from an old University of Adelaide Latex report template.

9 References

De Nicolò, V, P Villari, and C De Vito. 2022. "Impact of COVID-19 Vaccination on Risk Perception: A Cross-Sectional Study on Vaccinated People." *European Journal of Public Health* 32 (Supplement_3). https://doi.org/10.1093/eurpub/ckac129.664.

Eikenberry, Steffen E., Marina Mancuso, Enahoro Iboi, Tin Phan, Kendall Eikenberry, Yang Kuang, and Abba B. Gumel. 2020. "To Mask or Not to Mask: Modeling the Potential for Face Mask Use by the General Public to Curtail the COVID-19 Pandemic." *Infectious Disease Modelling* 5: 293–308. https://doi.org/10.1016/j.idm.2020.04.001.

[2] See "percentage of times waiting room exceeded" in Appendix D.4. This calculation was generated by the function `room_prob()` on lines 154 – 165 of the file `silmarils.R`. This function processes the data output from the `elvish_wait()` function on lines 13 – 22 of the same file.

Guo, Yu, Hongzhe Xiang, and Yuhan Wang. 2023. "Understanding Self-Protective Behaviors During COVID-19 Pandemic: Integrating the Theory of Planned Behavior and o-s–o-r Model." *Current Psychology* 43 (13): 12071–83. https://doi.org/10.1007/s12144-023-04352-3.

Haischer, Michael H., Renee Beilfuss, Mikaela R. Hart, Luis Opielinski, David Wrucke, Gretchen Zirgaitis, Taylor D. Uhrich, and Sandra K. Hunter. 2020. "Who Is Wearing a Mask? Gender-, Age-, and Location-Related Differences During the COVID-19 Pandemic." Edited by Jianhong Sun. *PLOS ONE* 15 (10): e0240785. https://doi.org/10.1371/journal.pone.0240785.

Rana, P. S., Kalpana, Chahat, Sumit Kumar Modi, Alok L. Yadav, and Sahil Singla. 2023. "Comparative Analysis of Tree-Based Models and Deep Learning Architectures for Tabular Data: Performance Disparities and Underlying Factors." In *2023 International Conference on Advanced Computing and Communication Technologies (ICACCTech)*, 224–31. https://doi.org/10.1109/ICACCTech61146.2023.00044.

Ryan, Michael, Jingjing Ye, Julian Sexton, Roslyn I. Hickson, and Emily Brindal. 2025. "Face Mask Mandates Alter Major Determinants of Adherence to Protective Health Behaviours in Australia." *Royal Society Open Science* 12 (3): 241941. https://doi.org/10.1098/rsos.241941.

10 Statistical appendix

This section should contain all the code and output that you need to justify the claims you made above. Only include the code that actually contributes to your Results or Methods sections (maybe also the Background section). We don't need to see every single line of code you wrote, just the important bits that are needed for the reader to be able to obtain your results. Again, this is another balancing act between being too long and irrelevant, or being too short and incomplete (which would open you up to claims of academic misconduct or negligence). Research is a tough business.

Also, don't just paste all the code in here. Use subheadings and text to say what the code is, and what it is used for.

10.1 Appendix A: First Appendix

This text is under a subheading.

A code chunk running R:

```
a <- 1
b <- a+2
b
```

```
## [1] 3
```

You can run python from an Rmd file. But make sure you have the `reticulate` package installed first. Then the following code will run:

```
import numpy
x = numpy.arange(0.0, 2.0, 0.5)
y = 1 + numpy.sin(numpy.pi*x)
y
```

```
## array([1., 2., 1., 0.])
```

The following code chunk doesn't actually run, it is just displaying the code. You can use this if you want to include some code which takes too many hours to actually compile. But then you need to say which file you actually used to generate the data (since this code is not actually doing anything).

```
a <- 1
b <- a+2
b
```

10.1.1 Appendix A.1: A subsubsection of Appendix A

This text is under a subsubheading.

10.2 Appendix B: The second Appendix

We've now started another Appendix.

10.2.1 Appendix B.1: A subsection of Appendix B

This text is under a subsubheading.

10.2.2 Appendix B.2: Another subsection of Appendix B

This text is under a subsubheading.

Figure 1: T-Rex picture. Source Wikipedia: https://commons.wikimedia.org/wiki/File:Tyrannosaurus_Rex_colored.png